

CHAPTER 2

BIOMECHANICS OF RESISTANCE EXERCISE

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After completing this chapter, you will be able to

- identify the major components of skeletal musculature;
- differentiate the various types of levers of the musculoskeletal system;
- identify primary anatomical movements during sport activities and exercises;
- outline core mechanical factors associated with movement;
- describe the factors contributing to human strength and power;
- evaluate resistive force and power patterns of exercise devices; and
- identify factors of importance for joint biomechanics with exercise.

Knowledge of biomechanics is important for describing and understanding human movements, including those involved in sport and exercise. **Biomechanics** focuses on the mechanisms through

which the components of the musculoskeletal system interact to create movement. Having insight into how body movements are carried out and the stresses that the movements place on the musculoskeletal system facilitates the design of safe and effective resistance training programs and maximizes the potential for human endurance, strength, and power.

This chapter begins with an overview of skeletal musculature, body mechanics, and the primary movement patterns during sport activities and exercise, followed by biomechanical principles that connect the manifestation of human strength and movement. Next, the primary sources of resistance to muscle contraction used in exercise devices—including gravity, inertia, friction, fluid resistance, and elasticity—are discussed. Then we turn to concerns with resistance training that relate to joint biomechanics (with special emphasis on the shoulders, back, and knees).

Skeletal Musculature

To cause movement or to generate **force** against external objects, both ends of each skeletal muscle must be attached to bone by connective tissue. Traditionally, anatomists define the muscle's **origin** as its **proximal** (toward the center of the body) attachment and its **insertion** as its **distal** (away from the center of the body) attachment. Sometimes the origin is defined as the more stationary structure to which the muscle is attached and the insertion as the more mobile structure. This definition can lead to a confusing reversal of the origin and insertion. For example, during a straight-leg sit-up, the origin of the iliacus muscle is the femur, because of its relative immobility. The pelvis, being more mobile, is the insertion. However, during the leg raise exercise, the pelvis is relatively immobile and would therefore become the origin, while the more mobile femur would become the insertion. The traditional definition therefore provides the most consistency.

Muscles are attached to bone in various ways. In **fleshy attachments**, which are most often found at the proximal end of a muscle, muscle fibers are directly affixed to the bone, usually over a wide area so that force is distributed rather than localized. **Fibrous attachments**, such as **tendons**, blend into and are continuous with both the muscle sheaths and the connective tissue surrounding the bone. They have additional fibers that extend into the bone itself, making for a very strong union.

Virtually all body movements involve the action of more than one muscle. The muscle most directly involved in bringing about a movement is called the *prime mover*, or **agonist**. A muscle that can slow down or stop the movement is called the **antagonist**. The antagonist assists in joint stabilization and in braking the limb toward the end of a fast movement, thereby protecting ligamentous and **cartilaginous joint** structures from potentially destructive forces. During throwing, for example, the triceps acts as an agonist, extending the elbow to accelerate the ball. As the elbow approaches

full extension, the biceps acts as an antagonist to slow down elbow extension and bring it to a stop, thereby protecting elbow structures from internal impact.

A muscle is called a **synergist** when it assists indirectly in a movement. For example, the muscles that stabilize the scapula act as synergists during upper arm movement. Without these synergists, the muscles that move the upper arm (many of which originate on the scapula) would not be effective in bringing about this movement. Synergists are also required to control body motion when the agonist is a muscle that crosses two joints. For example, the rectus femoris muscle crosses the hip and knee, acting to flex the hip and extend the knee when contracting. Rising from a low squat involves both hip and knee extension. If the rectus femoris is to act to extend the knee as a person rises without inclining the trunk forward, hip extensor muscles such as the gluteus maximus must act synergistically to counteract the hip flexion that would otherwise result from tension in the rectus femoris.

Levers of the Musculoskeletal System

Although there are many muscles in the body that do not act through levers, such as muscles of the face, tongue, heart, arteries, and sphincters, body movements directly involved in sport and exercise primarily act through the bony levers of the skeleton. In order to understand how the body affects such movements, a basic knowledge of levers is required. Several basic definitions follow.

first-class lever—A lever for which the muscle force and resistive force act on opposite sides of the fulcrum ([figure 2.1](#)).

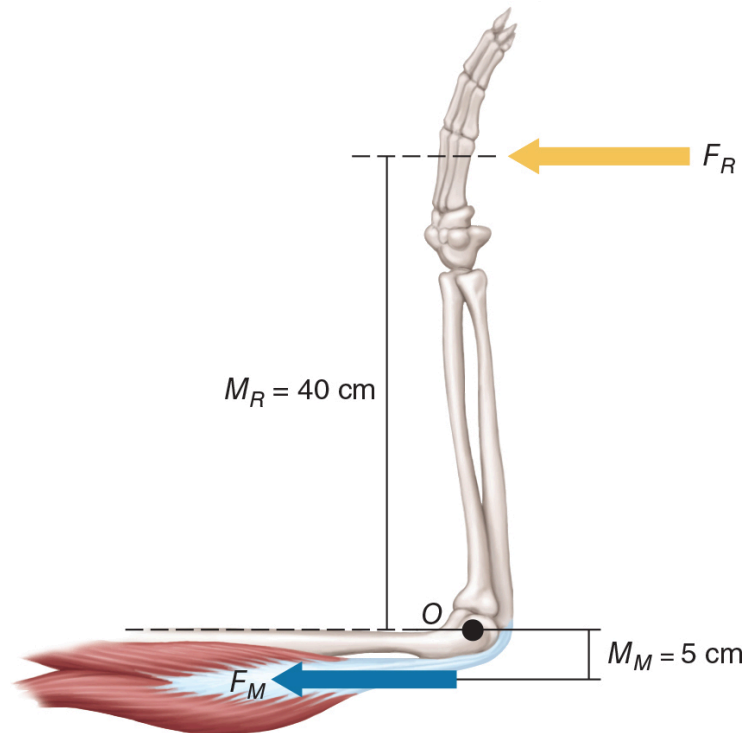


FIGURE 2.1 A first-class lever (the forearm). Elbow extension against resistance (e.g., a triceps extension exercise). O = fulcrum; F_M = muscle force; F_R = resistive force; M_M = moment arm of the muscle force; M_R = moment arm of the resistive force. Mechanical advantage = $M_M/M_R = 5 \text{ cm}/40 \text{ cm} = 0.125$, which, being less than 1.0, is a disadvantage in the common sense.

fulcrum—The pivot point of a lever.

lever—A rigid or semirigid body that, when subjected to a force whose line of action does not pass through its pivot point, exerts force on any object impeding its tendency to rotate ([figure 2.2](#)).

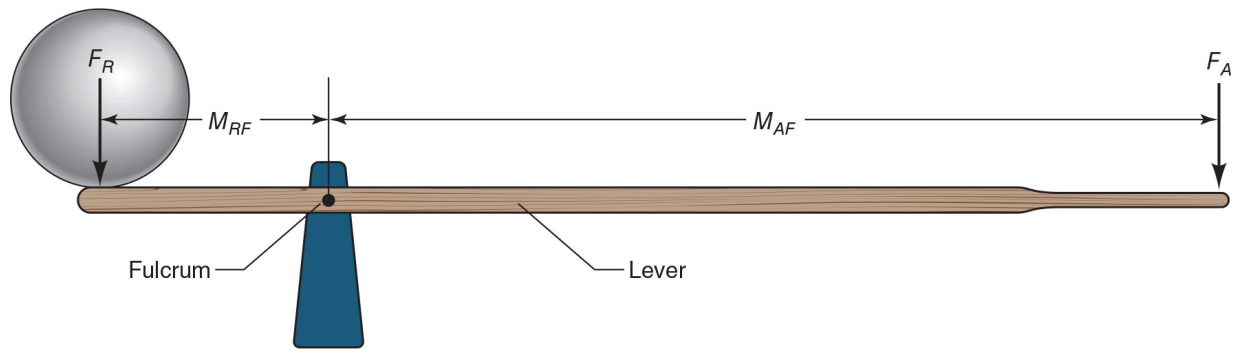


FIGURE 2.2 A lever. The lever can transmit force tangential to the arc of rotation from one contact point along the object's length to another. F_A = force applied to the lever; M_{AF} = moment arm of the applied force; F_R = force resisting the lever's rotation; M_{RF} = moment arm of the resistive force. The lever applies a force on the object equal in magnitude to but opposite in direction from F_R .

mechanical advantage—The ratio of the moment arm through which an applied force acts to that through which a resistive force acts. For there to be a state of equilibrium between the applied and resistive torques, the product of the muscle force and the moment arm through which it acts must equal the product of the resistive force and the moment arm through which it acts. Therefore, a mechanical advantage, represented as a ratio greater than 1.0, allows the applied (muscle) force to be less than the resistive force to produce an equal amount of torque. Conversely, a calculated mechanical advantage of a ratio less than 1.0 indicates that one must apply greater (muscle) force than the amount of resistive force present, creating an obvious disadvantage for the muscle.

moment arm (also called *force arm*, *lever arm*, or *torque arm*)—The perpendicular distance from the line of action of the force to the fulcrum. The line of action of a force is an infinitely long line passing through the point of application of the force, oriented in the direction in which the force is exerted.

muscle force—Force generated by biochemical activity, or the stretching of noncontractile tissue, which tends to draw the opposite ends of a muscle toward each other.

resistive force—Force generated by a source external to the body (e.g., gravity, inertia, friction) that acts contrary to muscle force.

second-class lever—A lever for which the muscle force and resistive force act on the same side of the fulcrum, with the muscle force acting through a moment arm longer than that through which the resistive force acts, as when the calf muscles work to raise the body onto the balls of the feet ([figure 2.3](#)). Due to its mechanical advantage (i.e., relatively long moment arm), the required muscle force is smaller than the resistive force (body weight).

third-class lever—A lever for which the muscle force and resistive force act on the same side of the fulcrum, with the muscle force acting through a moment arm shorter than that through which the resistive force acts ([figure 2.4](#)). The mechanical advantage is thus less than 1.0, so the muscle force has to be greater than the resistive force to produce torque equal to that produced by the resistive force.

torque (also called **moment**)—The degree to which a force tends to rotate an object about a specified fulcrum. It is defined quantitatively as the magnitude of a force times the length of its moment arm.

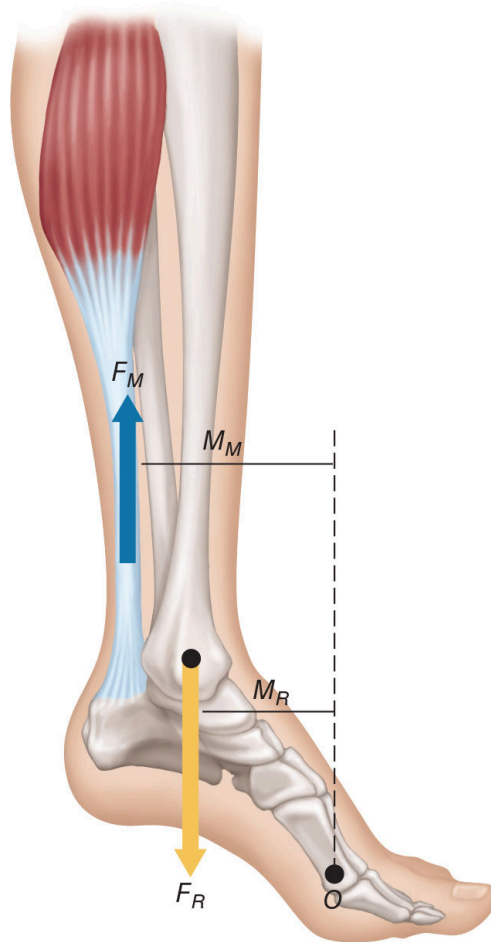


FIGURE 2.3 A second-class lever (the foot). Plantar flexion against resistance (e.g., a standing heel raise exercise). F_M = muscle force; F_R = resistive force; M_M = moment arm of the muscle force; M_R = moment arm of the resistive force. When the body is raised, the ball of the foot, being the point about which the foot rotates, is the fulcrum (O). Because M_M is greater than M_R , F_M is less than F_R .

[Figure 2.1](#) shows a first-class lever, because muscle force and resistive force act on opposite sides of the fulcrum. During isometric exertion or constant-speed joint rotation, $F_M \times M_M = F_R \times M_R$. Because M_M is much smaller than M_R , F_M must be much greater than F_R ; this illustrates the disadvantageous nature of this arrangement (i.e., a large muscle force is required to push against a relatively small external resistance).

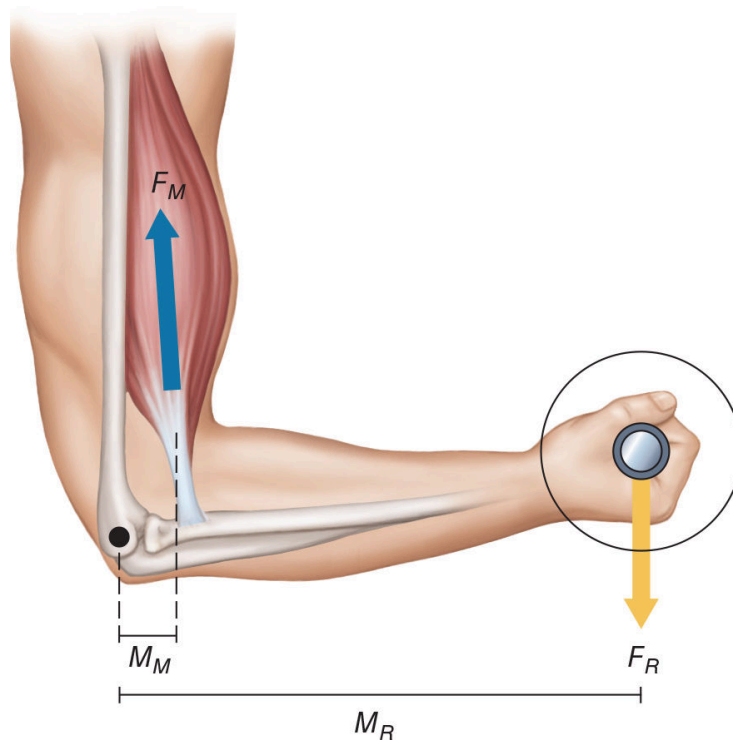
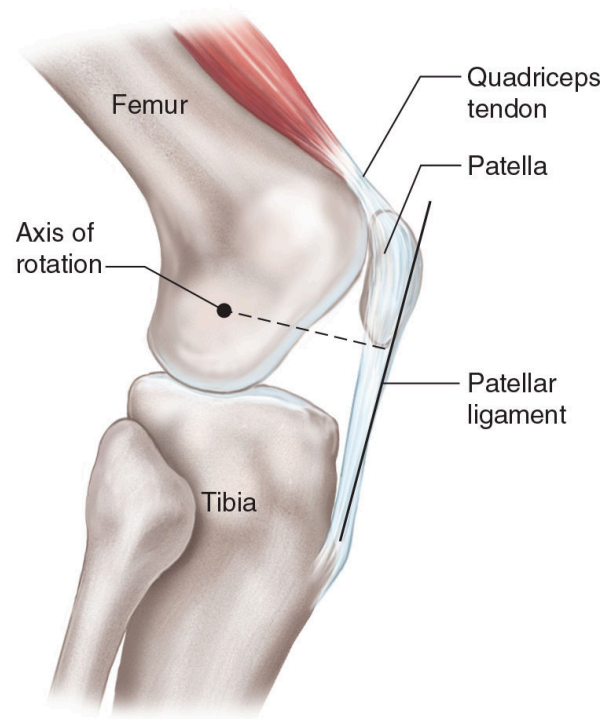


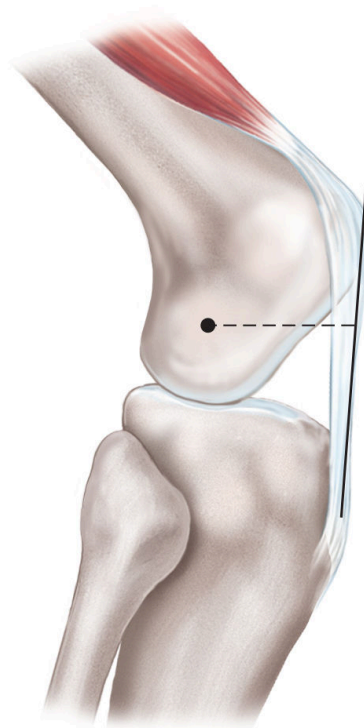
FIGURE 2.4 A third-class lever (the forearm). Elbow flexion against resistance (e.g., a biceps curl exercise). F_M = muscle force; F_R = resistive force; M_M = moment arm of the muscle force; M_R = moment arm of the resistive force. Because M_M is much smaller than M_R , F_M must be much greater than F_R .

Most human muscles that rotate the limbs about body joints operate at a mechanical advantage of less than 1.0 (that is, at a mechanical *disadvantage*). This is why internal muscle forces are much greater than the forces exerted by the body on external objects. For example, in [figure 2.2](#), because the resistance moment arm is eight times longer than the muscle moment arm, muscle force must be eight times the resistive force. The extremely high internal forces experienced by muscles and tendons account in large part for injury to these tissues. During actual movement, the categorization of a lever as first, second, or third class often depends on the somewhat arbitrary decision of where the fulcrum lies. Therefore, understanding the principle of mechanical advantage is of much greater importance than being able to classify levers.

Mechanical advantage often changes continuously during real-world activities. The following are examples of this.



a



b

FIGURE 2.5 Mechanical advantage at the knee. (a) The patella increases the mechanical advantage of the quadriceps muscle group by maintaining the quadriceps tendon's distance from the knee's axis of rotation. (b) Absence of the patella allows the tendon to fall closer

to the knee's center of rotation, shortening the moment arm through which the muscle force acts and thereby reducing the muscle's mechanical advantage.

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- For movements such as knee extension and flexion, where the joint is not a true hinge, the location of the axis of rotation changes continuously throughout the range of motion, affecting the length of the moment arm through which the quadriceps and hamstrings act. For knee extension, the patella, or kneecap, helps to prevent large changes in the mechanical advantage of the quadriceps muscle by keeping the quadriceps tendon from falling in close to the axis of rotation ([figure 2.5](#); 17).
- For movements such as elbow extension and flexion, there is no structure such as the patella to keep the perpendicular distance from the joint axis of rotation to the tendon's line of action relatively constant ([figure 2.6](#)).
- During resistance training with free weights, the moment arm through which the weight acts equals the horizontal distance from a line through the center of mass of the barbell or dumbbell to the body joint about which rotation of the limb occurs; the resistive moment arm thus varies throughout the movement ([figure 2.7](#)).

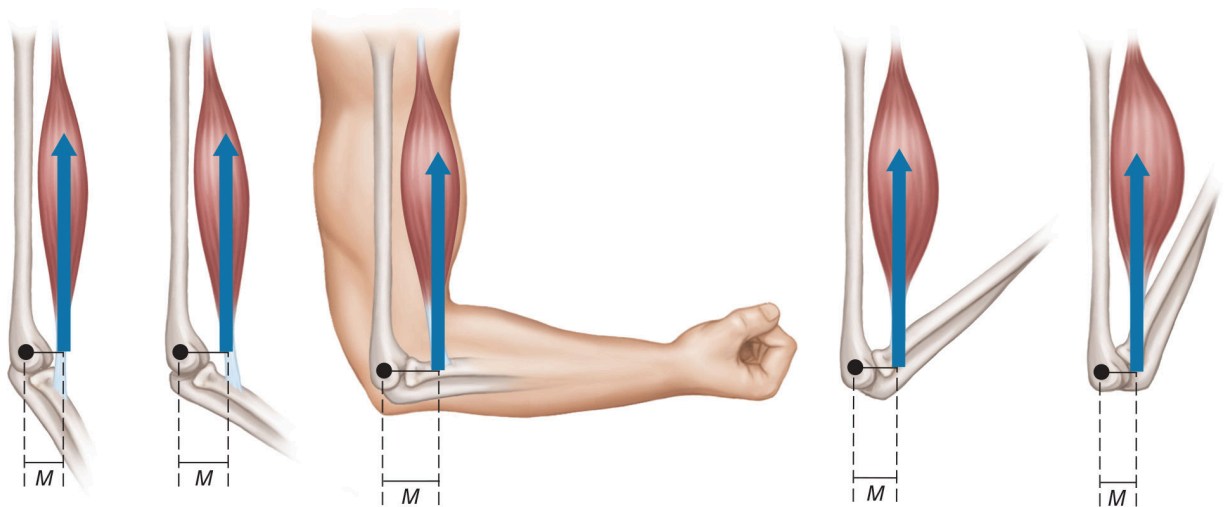


FIGURE 2.6 Mechanical advantage at the elbow. During elbow flexion with the biceps muscle, the perpendicular distance from the joint axis of rotation to the tendon's line of action varies throughout the range of joint motion. When the moment arm (M) is shorter, there is less mechanical advantage.

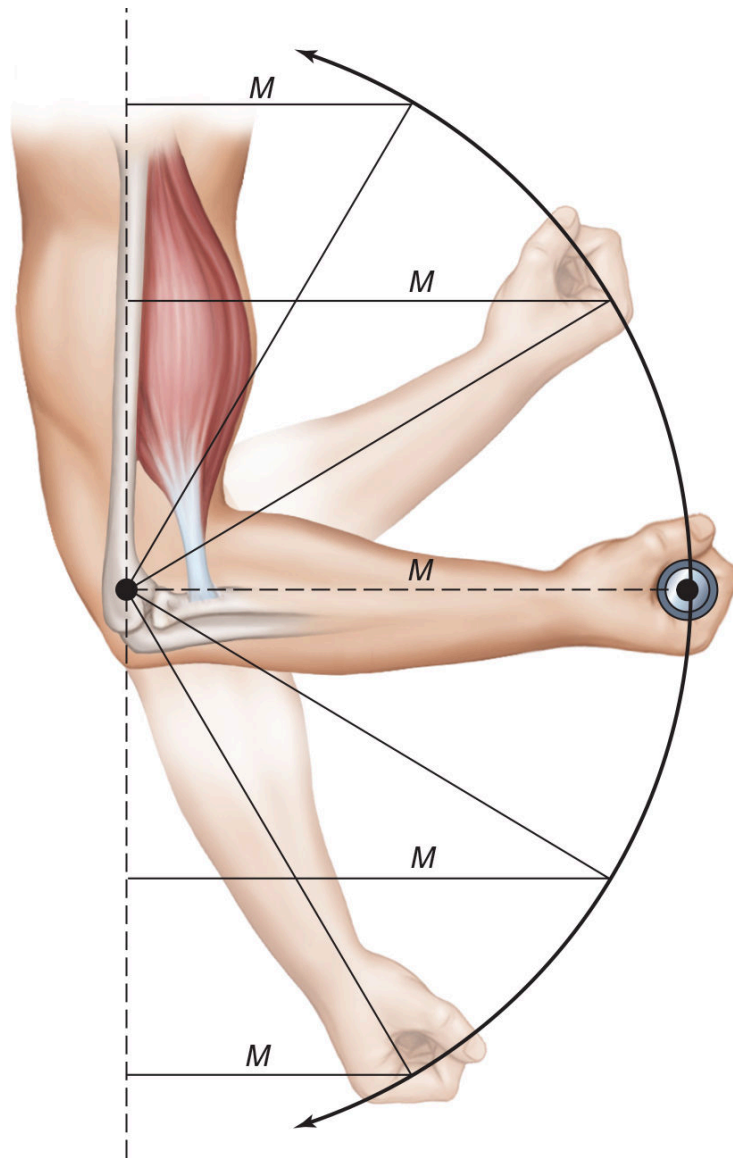


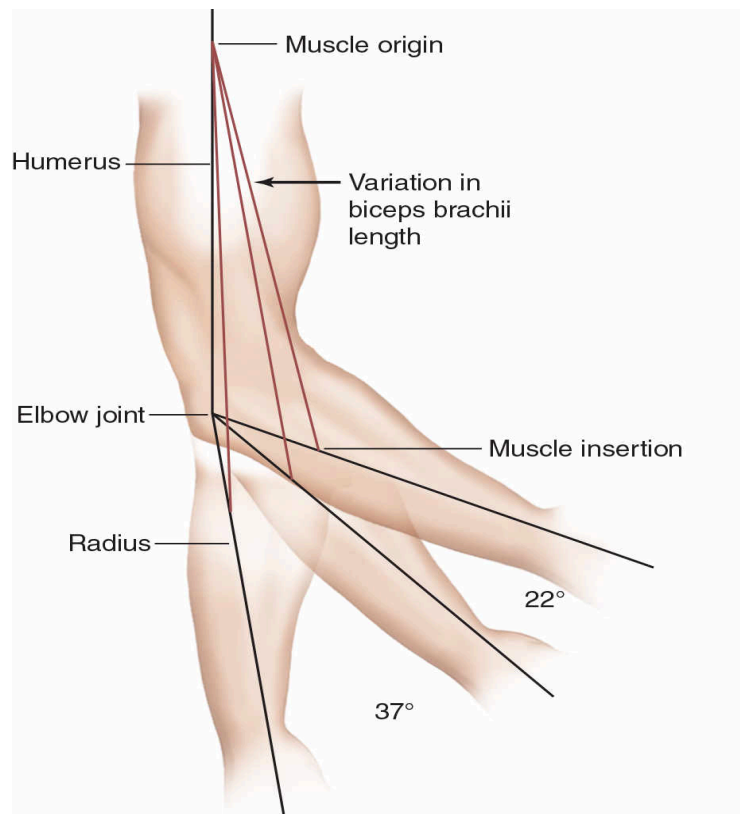
FIGURE 2.7 As a free weight is lifted, the moment arm (M) through which the weight acts, and thus the resistive torque, changes with the horizontal distance from the weight to the elbow.

Most of the skeletal muscles operate at a considerable mechanical disadvantage due to the lever arrangement within the body and relative to the external forces the body resists. Thus, during sport and other physical activities, forces in the muscles and tendons are much

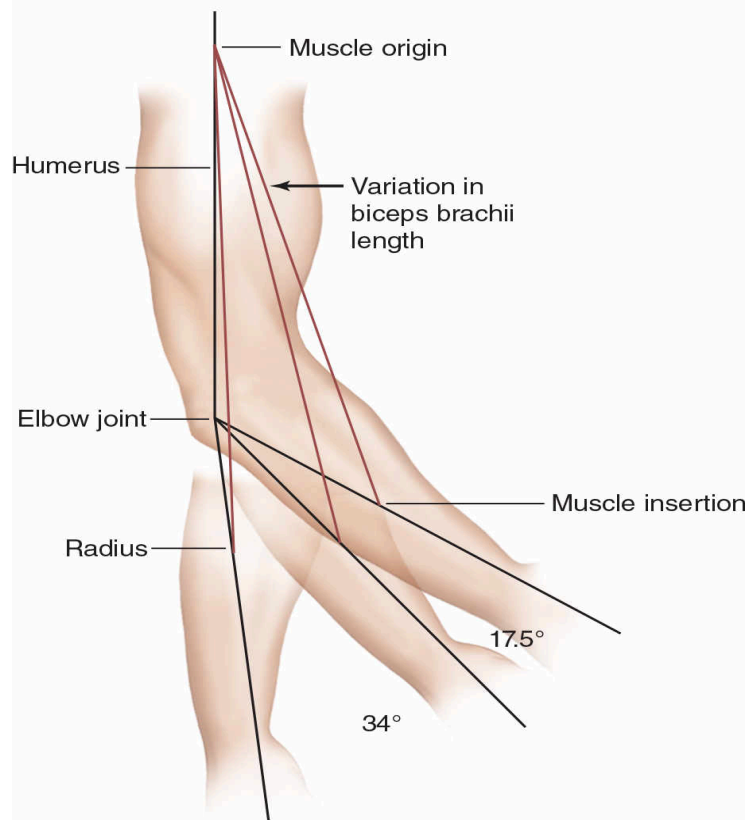
higher than those exerted by the hands or feet on external objects or the ground.

Variations in Tendon Insertion

Considerable variation in human anatomical structure exists, including the points at which tendons are attached to bone. A person whose tendons are inserted on the bone farther from the joint center should be able to lift heavier weights because muscle force acts through a longer moment arm and thus can produce greater torque around the joint. (In [figure 2.6](#), for example, consider how the moment arm $[M]$ would change if the tendon insertion were farther to the right.) It is important, however, to recognize the trade-off involved in tendon insertion. The mechanical advantage gained by having tendons insert farther from the joint center is accompanied by a loss of maximum speed because, with the tendon inserted farther from the joint center, the muscle has to contract more to make the joint move through a given range of motion. In other words, a given amount of muscle shortening results in less rotation of body segments about a joint, which translates into a loss in movement speed. [Figure 2.8a](#) shows that, starting with the joint extended, when a hypothetical muscle shortens by a given amount, the joint rotates by 37 degrees. However, if the muscle were inserted farther from the joint center, as in [figure 2.8b](#), the same amount of muscle shortening would bring about only 34 degrees of joint rotation because of the geometry of the dynamic triangle whose vertices are the muscle insertion and origin and the joint center of rotation.



a



b

FIGURE 2.8 Changes in joint angle with equal increments of muscle shortening when the tendon is inserted (*a*) closer to and (*b*) farther from the joint center. Configuration *b* has a larger moment arm and thus greater torque for a given muscle force but less rotation per unit of muscle contraction and thus slower movement speed.

To produce a given joint rotational velocity, a muscle inserted farther from the joint center must contract at a higher speed, at which it can generate less force due to the inverse force–velocity relationship of muscle (45, 46) described later in this chapter. Therefore, such a tendon arrangement reduces the muscle’s force capability during faster movements.

One can see how relatively subtle individual differences in structure can result in various advantages and disadvantages. Although these skeletal arrangements are nonmodifiable, it is important to understand that for slower movements, as in powerlifting, tendon insertion farther from the joint than normal can be advantageous, while for athletic activities occurring at high speeds, such as hitting a tennis ball, such an arrangement can be disadvantageous.

Anatomical Planes and Major Body Movements

[Figure 2.9a](#) illustrates a person standing in the standard **anatomical position**, standing upright, arms by the sides, and with the palms facing forward. So-called anatomical views of the body are often used in clinical settings (e.g., medical imaging), but in biomechanics they also provide reference positions that can be used when describing human movement. The **sagittal**, **frontal**, and **transverse planes** present the body as if sliced into left–right, front–back, and upper–lower sections, respectively, although not necessarily through the midpoint. Examples of exercise movements that take place in these three planes include the biceps curl (sagittal; [figure 2.9b](#)), lateral raise (frontal; [figure 2.9c](#)), and dumbbell fly (transverse; [figure 2.9d](#)).

Biomechanical analysis of human movement can be used to select and quantitatively analyze the target activity. Depending on its complexity, often simple visual observation is adequate for identifying the basic features of the movement pattern and the opportunities for developing or maintaining postural form to achieve the desired performance outcome. Exercises can be selected that target a similar plane of movement around the same joints, thereby promoting specificity of training. Then with practice, this approach can be extended to compound exercises, which concurrently use multiple planes of movement.

[Figure 2.10](#) presents a grid of body movements that could be used to develop a programming approach oriented to specific anatomical planes. Selecting exercises for which the dominant plane is either sagittal, frontal, or transverse might seem restrictive, but since few body movements occur in one of these major planes in isolation, there is usually a cross-training effect; that is, exercising muscles within the planes also strengthens them for movements between the planes. Although a program providing resistance

exercises for all the movements in [figure 2.10](#) would be both comprehensive and balanced, some of the movements are commonly omitted from standard exercise programs whereas others receive particular emphasis.

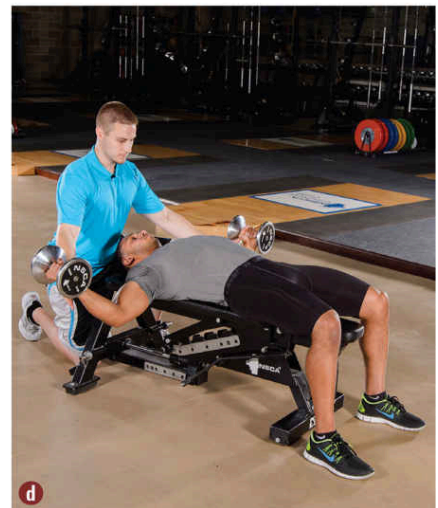
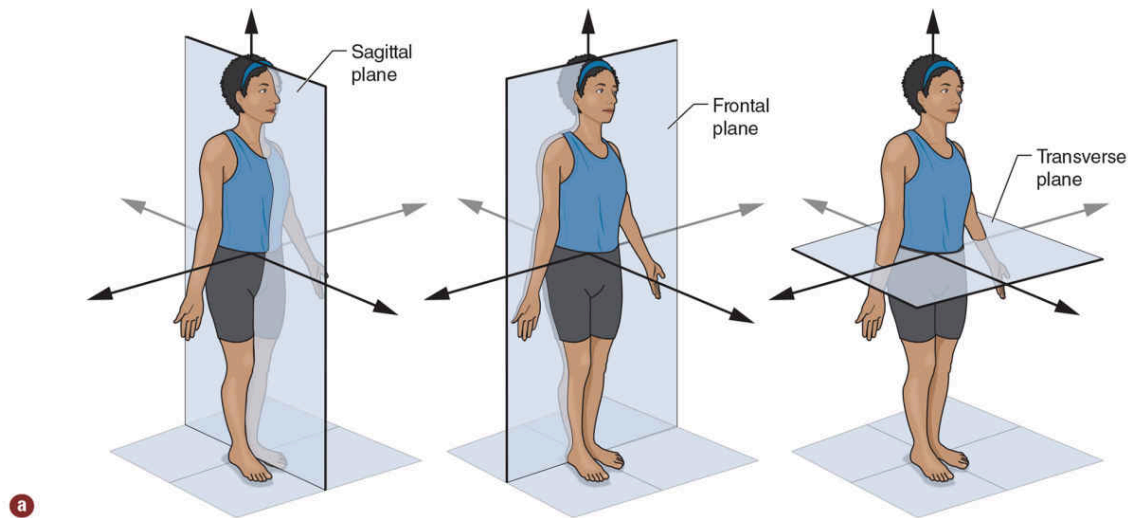


FIGURE 2.9 The three planes of the human body in the (a) anatomical position with exercise examples: (b) biceps curl (sagittal), (c) lateral raise (frontal), and (d) dumbbell fly (transverse).

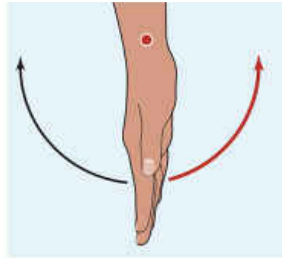
Standard resistance training programs often lack particular sport-specific exercises, such as shoulder internal and external rotation (used in throwing, serving, pitching, or bowling), knee flexion

(sprinting), hip flexion (kicking, sprinting), ankle dorsiflexion (running), hip internal and external rotation (landing, pivoting), hip adduction and abduction (lateral cutting or changing direction), and torso transverse rotation (throwing, hitting, or batting). Incorporating the principle of dynamic correspondence into training programs can enhance their effectiveness for sport performance. This principle involves elements such as the amplitude and direction of movements, accentuated regions of force production, dynamics of effort, rate and timing of maximum force production, regime of muscular work, and segmental interrelation (16, 55). By considering these criteria and accounting for the multiplanar contributions of muscles, joints, and body segments, strength and conditioning professionals can create exercise programs that more effectively target specific sporting activities, thereby maximizing their athletes' chances of success.

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Flexion, reduce angle between palm of hand and inner forearm
Exercise: wrist curl; wrist roller
Sport: basketball free throw; volleyball serve

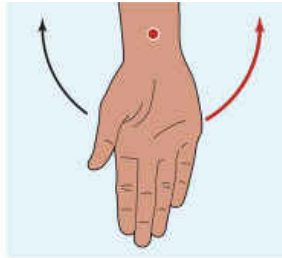
Extension, increase angle between palm of hand and inner forearm
Exercise: wrist extension; wrist roller
Sport: tennis backhand; gymnastics handstand



2. Wrist—frontal

Ulnar deviation, reduce angle between hand and ulna bone (pinky side of forearm)
Exercise: battle rope (pull down phase)
Sport: baseball hitting swing; tennis forehand stroke

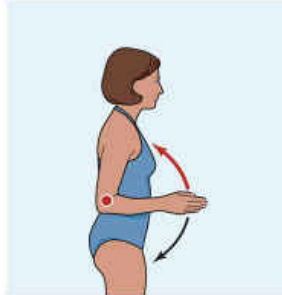
Radial deviation, reduce angle between hand and radius bone (thumb side of forearm)
Exercise: battle rope (lifting phase)
Sport: golf backswing; tennis backhand



3. Elbow—sagittal

Flexion, reduce angle between forearm and upper arm
Exercise: biceps curl
Sport: rowing; rock climbing; boxing uppercut punch

Extension, increase angle between lower arm and upper arm
Exercise: triceps pushdown; wheelchair push-up
Sport: shot put; American football quarterback throw



4. Forearm—frontal

Pronation, inward forearm rotation (palm facing posterior, rear)

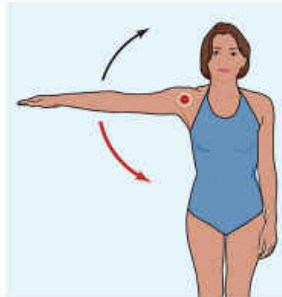
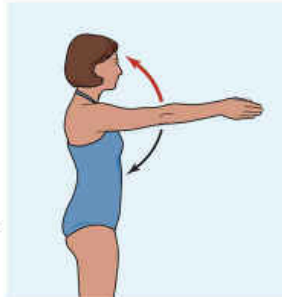
Supination, outward forearm rotation (palm facing anterior, front)

Exercise: Arnold press (starting forearm pronation to supination overhead)

5. Shoulder—sagittal

Flexion, increase angle between arm and body by raising in front
Exercise: front shoulder raise
Sport: baseball pitching (early phase); swimming backstroke

Extension, reduce angle between arm and body by lowering from raised position
Exercise: dumbbell pullover (concentric phase)
Sport: freestyle swimming stroke; cricket bowling



6. Shoulder—frontal

Adduction, reduce angle between inner arm and outer hip
Exercise: wide grip lat pulldown (upper arm)
Sport: swimming breaststroke; gymnastics rings (lateral push down)

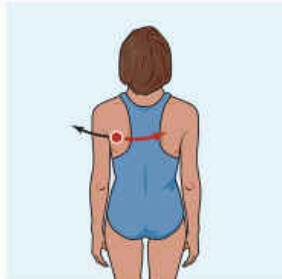
Abduction, increase angle between inner arm and outer hip
Exercise: dumbbell lateral raise
Sport: yoga sun salutation (surya namaskar); springboard diving (both in reach-up phase); weightlifting snatch

7. Shoulder—frontal

Elevation, superior scapula movement (toward head)

Depression, inferior scapula movement (toward feet)

Exercise: trap raise (shoulder shrug); scapula pull-up
Sport: swimming freestyle stroke; gymnastics L-sit



8. Shoulder—transverse

Retraction, (scapula adduction), medial scapular movement (toward midline)

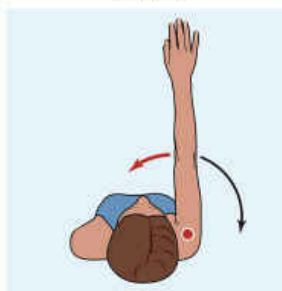
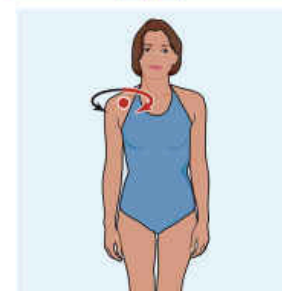
Protraction (scapula abduction), lateral scapular movement (away from midline)

Exercise: high rope pull; scapula push-up
Sport: archery; boxing

9. Shoulder—transverse

Internal rotation, (e.g., elbow flexed to 90°) reduce angle between lower arm and trunk
Exercise: shoulder internal rotation
Sport: baseball pitching throw; javelin throw

External rotation, (e.g., elbow flexed to 90°) increase angle between lower arm and trunk
Exercise: shoulder external rotation
Sport: karate block; baseball pitching layback



10. Shoulder—transverse

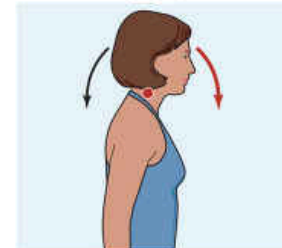
Adduction, upper arm horizontal, 90° to trunk, reduce angle between arm and chest
Exercise: seated row (concentric phase)
Sport: tennis forehand; boxing cross punch

Abduction, upper arm horizontal, 90° to trunk, increase angle between arm and chest
Exercise: bench press (eccentric phase)
Sport: tennis backhand; swimming backstroke

11. Neck—sagittal

Flexion, reduce angle between chin and chest
Exercise: neck machine
Sport: gymnastics somersault; soccer heading ball; rugby scrum

Extension, increase angle between chin and chest
Exercise: dynamic back bridge
Sport: gymnastics back flip; diving; wrestling



12. Neck—transverse

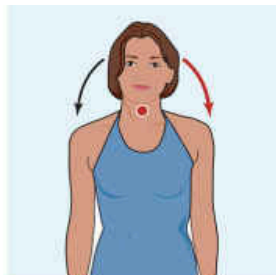
Left/right rotation, head upright, rotate head to reduce angle between chin and shoulder
Exercise: manual resistance
Sport: wrestling movement; tennis (ball tracking); motor racing (head stability)

FIGURE 2.10 Major body movements. Planes of movement are relative to the body in the anatomical position unless otherwise stated. Common exercises that provide resistance to the movements and related sport activities are listed.

13. Neck—frontal

Left/right tilt, reduce angle between ear and shoulder

Exercise: neck machine;
band resistance
Sport: rugby scrum and tackle;
slalom skiing; motor racing
(head stability)

**15. Lower back—frontal**

Left/right tilt, reduce angle between outer upper and lower body

Exercise: side bend; medicine ball overhead hook throw
Sport: gymnastics side aerial;
basketball hook shot; shot put
(load phase)

**17. Hip—sagittal**

Flexion, reduce angle between thigh and trunk

Exercise: leg raise; back squat eccentric phase (with knee flexion)
Sport: American football punt;
soccer kick

Extension, increase angle between thigh and trunk

Exercise: back squat (concentric phase); glute bridge
Sport: long jump take-off; sprinting;
weightlifting second pull

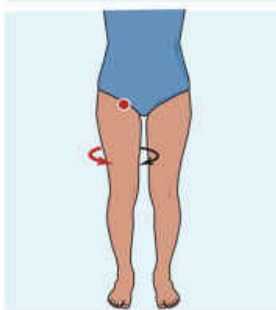
**19. Hip—transverse**

Internal rotation, rotate upper leg toward midline without upper body contribution

Exercise: 90-90 drill (figure four pose)
Sport: basketball pivot; golf lead leg (swing phase)

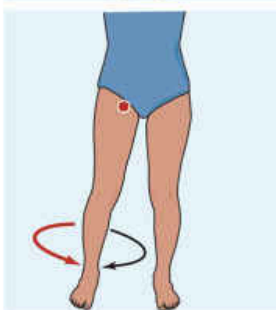
External rotation, rotate upper leg away from midline without upper body contribution

Exercise: 90-90 drill (figure four pose)
Sport: figure skating turn; baseball pitching rear leg (landing phase)

**21. Hip—transverse**

Circumduction, circular motion of distal end of body segment

Exercise: hip circle
Sport: figure skating; kickboxing

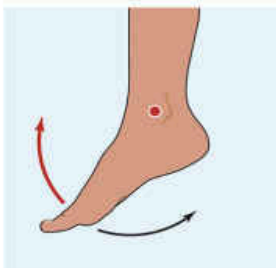
**23. Ankle—sagittal**

Dorsiflexion, reduce angle between foot and shin

Exercise: toe raise; band resisted dorsiflexion
Sport: baseball hitting (load phase);
gymnastics (dismount landing)

Plantar flexion, increase angle between foot and shin

Exercise: calf (heel) raise; weightlifting second pull (triple extension)
Sport: basketball (jump concentric phase); gymnastics (take-off)

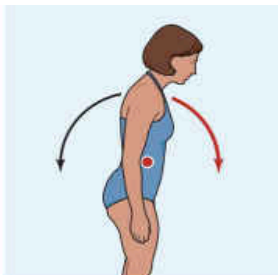
**14. Lower back—sagittal**

Flexion, reduce angle between chest and upper legs

Exercise: good morning (eccentric phase)
Sport: javelin throw (follow through);
rowing (catch phase)

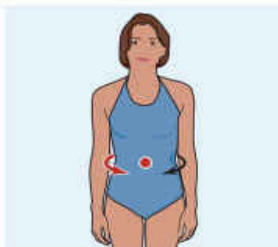
Extension, increase angle between chest and upper thighs

Exercise: stiff-leg deadlift
Sport: gymnastics back flip;
rowing (finish phase)

**16. Lower back—transverse**

Left/right rotation, changing angle between upper and lower body

Exercise: medicine ball side toss;
bird dog (anti-rotation); Russian twist
Sport: baseball hitting; golf swing;
discus throw

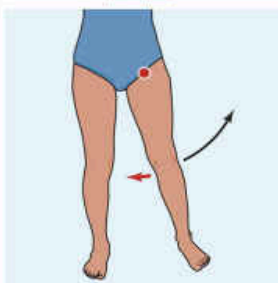
**18. Hip—frontal**

Adduction

Exercise: standing hip adduction machine; Copenhagen plank
Sport: soccer inside pass; figure skating crossover step

Abduction

Exercise: standing hip abduction machine; cable stack abduction;
Cossack squat (eccentric phase)
Sport: rollerblading; speed ice skating

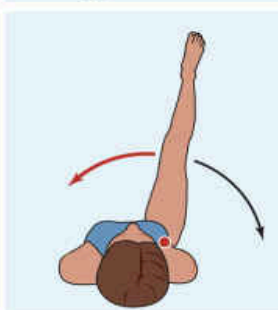
**20. Hip—transverse**

Adduction, thigh horizontal, 90° to trunk, rotate leg toward midline

Exercise: seated hip adduction machine; seated band hip adduction
Sport: karate in-sweep (uchi-barai);
figure skating jump landing stabilization

Abduction, thigh horizontal, 90° to trunk, rotate leg away from midline

Exercise: seated hip abduction machine;
seated band hip abduction
Sport: wrestling escape maneuver;
rugby tackle side-step

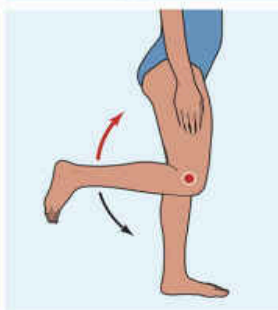
**22. Knee—sagittal**

Flexion, reduce angle between lower and upper leg

Exercise: leg (knee) curl; back squat eccentric phase
Sport: sprinting (swing forward phase);
basketball (jump landing phase)

Extension, increase angle between lower and upper leg

Exercise: leg (knee) extension; back squat concentric phase
Sport: sprinting (push-off phase);
volleyball spiking

**24. Ankle—frontal**

Inversion, rotate foot (big toe, hallux) toward midline of body

Exercise: band resisted ankle inversion;
progressive landing drill
Sport: soccer dribbling; snowboarding

Eversion, rotate foot (fifth toe) outward away from midline of body

Exercise: band-resisted ankle eversion;
progressive landing drill
Sport: basketball defensive maneuver;
soccer change of direction

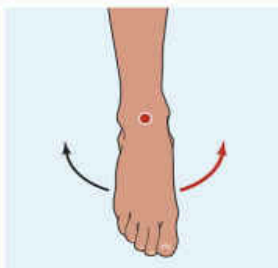


FIGURE 2.10 *(continued)*

Human Strength and Power

The terms *strength* and *power* are widely used to describe some important abilities that contribute to maximal human efforts in sport and other physical activities. Unfortunately, there is often little consistency in the way the terms are used. This section provides a scientific basis for understanding human strength and power and shows how various factors contribute to their manifestation.

Basic Definitions

Though it is widely accepted that **strength** is the ability to exert force, there is considerable disagreement as to how strength should be measured. The load that a person can lift is probably the oldest quantitative measure of strength. Technological developments have popularized the use of isometric strength testing and also isokinetic strength testing. All sports involve **acceleration** (the change in **velocity** with respect to time) of the body, a sporting implement (e.g., baseball bat, javelin, tennis racket), or both. Acceleration is associated with resistive force according to Isaac Newton's second law of motion:

$$F = m \times a \quad (2.1)$$

Where F refers to force, m refers to mass, and a refers to acceleration. For clarity, force can be thought of as how hard we push or pull something, while **mass** refers to the stuff (e.g., bone, muscle, and organs) that we are made of. Although it can seem abstract, the impulse–momentum relationship, an extension of equation 2.1, provides more context (equation 2.2).

$$F \times \Delta t = m \times \Delta v \quad (2.2)$$

where everything to the left of the equal sign describes **impulse** and everything to the right of the equal sign describes **momentum** (more specifically, a *change* in momentum). This means that to cause (or control) the movement of an object (i.e., the mass of the

athlete, implement, or both), a certain amount of force has to be applied over the available time.

Here we have a cause and effect. For example, we know that vertical jump height is dictated by the vertical velocity that the jumper achieves at takeoff. Therefore, if a jumper with a mass of 74 kilograms (the athlete's weight is $74 \text{ kg} \times \text{gravity} = 725.94 \text{ newtons}$) wants to jump 40 centimeters, the athlete will have to take off at a vertical velocity of 2.82 m/s. This means that the athlete will have to apply an impulse of 210 Ns (newton-seconds). There is no way around this. If the athlete cannot achieve the necessary 210 Ns impulse, the athlete will not achieve the desired jump height. How the athlete achieves this impulse is another matter, though, because, as equation 2.2 shows, impulse is the product of net force (force not including body weight) and time.

To illustrate this point, the athlete could achieve the same jump height by applying a mean net force of 1,000 newtons over 0.210 seconds, 800 newtons over 0.263 seconds, 600 newtons over 0.350 seconds, or 400 newtons over 0.525 seconds. Of course, we would ideally like the athlete to get to a point where the athlete can apply as much force as possible so as to perform the vertical jump (or any other movement of interest) as quickly as possible. Finally, both impulse and momentum are **vector** quantities, which means that the athlete considers both magnitude (how much) and direction (which way); this is critical to sport performance. With this in mind, the strength and conditioning professional can use this information to develop testing protocols to establish the athlete's capacity to apply force over different time periods or against different loads. This will help the strength and conditioning professional to get a more complete picture of whether the athlete is able to use current strength levels and to assess whether they change because of sporting demands or training. Another way that strength and conditioning professionals have used biomechanical concepts to assess the athlete's capacity to apply force over different time

periods and against different loads has been with work and power (8).

The Impulse–Momentum Relationship

Male rugby athletes can jump between 0.37 and 0.44 meters (1.21-1.44 ft) (42). To jump 0.37 meters, they need to leave the ground at 2.694 m/s. Impulse (force $\times$ time) changes momentum (mass $\times$ velocity). Because mass does not change, only velocity needs to change. To do this at 95.2 kilograms (209.9 lb), they need to apply 256.6 newton-seconds of impulse. To do this from a propulsion phase of 0.232 seconds means applying a force 1105.6 newtons over their body weight. To jump 0.44 meters (16% more), heavier rugby athletes (98.2 kg [216.5 lb]) must leave the ground at 2.938 m/s, which requires 288.5 newton-seconds. They achieve this by applying 7% less force over 17% more time.

Positive Work and Power

When we apply force to an object that is moving, we are performing **work**. In a biomechanical context, work can be understood in a simpler form as the product of force and **displacement** (how far the object moves). This is often how work is described in resistance training and is familiar to many athletes and strength and conditioning professionals.

However, work can also be understood as the change in the mechanical energy of an object. **Mechanical energy** is the sum of **potential energy** and **kinetic energy** (equations 2.3 and 2.4):

$$\text{Potential energy} = m \times g \times h \quad (2.3)$$

where m is mass (as described above), g is the acceleration due to gravity ($9.81 \text{ m} \cdot \text{s}^{-2}$), and h is the change in height (or position). This represents the energy an object has because of its position.

$$\text{Kinetic energy} = \frac{1}{2} m \times v^2 \quad (2.4)$$

where m is mass and v is velocity. This represents the energy an object has due to its movement.

Together, these two forms of energy determine the total mechanical energy change, which is key to understanding the work

done in dynamic movements such as jumping or sprinting.

Finally, *power* refers to how quickly work is performed. In its simplest form, power is the amount of work divided by the time over which the force is applied. Athletes in many sports aim to maximize power output, meaning they strive to apply as much force as possible over the shortest time frame to achieve the required movement.

For all the equations in this chapter to work out correctly, consistent units must be used. In the International System of Units (SI, abbreviated from French), the worldwide standard, force is measured in newtons (N), length (e.g., distance and displacement) in meters (m), work in joules (J, i.e., newton-meters, or N · m), time in seconds (s), and power in watts (W, i.e., J/s). The appropriate SI units for the equations can be obtained from other common units using the factors listed in [table 2.1](#) (47a).

TABLE 2.1 Units of Measure and Conversions

Quantity	SI (international) unit	Conversion multiplier	Traditional (British) unit	Conversion multiplier	SI (international) unit
Distance	meter	× 3.28 =	foot	× 0.3048 =	meter
Distance	kilometer	× 0.621 =	mile	× 1.61 =	kilometer
Angle	radian	× 57.3 =	degree	× 0.0175 =	radian
Velocity	meters/second	× 2.24 =	miles/hour	× 0.447 =	meters/second
Velocity	kilometers/hour	× 0.621 =	miles/hour	× 1.609 =	kilometers/hour
Force	newton	× 0.225 =	pound	× 4.45 =	newton
Force	kilogram	× 2.205 =	pound	× 0.4535 =	kilogram
Work, energy	joule	× 0.738 =	foot-pound	× 1.356 =	joule
Power	watt	× 0.0013 =	horsepower	× 745.7 =	watt

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Quantity	SI (international) unit	Conversion multiplier	Traditional (British) unit	Conversion multiplier	SI (international) unit
Torque	newton-meter	× 0.738 =	foot-pound	× 1.356 =	newton-meter

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We can measure the velocity of the barbell with measurement devices such as accelerometers (measures the acceleration of the object from which velocity can be calculated) and linear position transducers (devices that have a cable that is extracted as the barbell is lifted upward, and retracts as it is lowered, recording how far and how fast this happens). Consider a set of standing overhead barbell presses with 60 kilograms for which we have velocity data. How can we calculate the work performed to move the barbell? The displacement of the barbell during the first repetition was 0.541 meters. We take this and multiply it by the mass of the barbell and then multiply this by gravity using equation 2.3 like so:

$$60 \text{ kg} \times 9.81 \text{ m} \cdot \text{s}^{-2} \times 0.541 \text{ m} = 318.43 \text{ J} \quad (2.5)$$

Or we could calculate potential energy and kinetic energy on a sample-by-sample basis. The linear position transducer that was used to collect the barbell press velocity data records at a sampling frequency of 1,000 hertz (Hz, or samples per second). We then add these together and calculate the change, which provides the same result as the one shown in equation 2.5. What happens if the barbell mass is increased or it is lifted over a greater range of motion? If we add 20 kilograms to the barbell and lift it over the same range of movement, the work performed on it increases to 424.58 joules. To achieve that amount of work with the same load on the barbell, the athlete would have to lift it over approximately 33% farther (0.721 m vs. 0.541 m). If this athlete performs multiple repetitions during a set and, as should be the case, the range of movement remains fairly constant, we can multiply the work performed on the barbell

(as calculated using equation 2.3) by the number of repetitions to calculate the amount of work performed during the set. We can do this over all sets of this exercise or all sets of all exercises for which we are able to calculate work with a degree of accuracy, and we can use this to provide an estimate of workload.

Power is the rate that work is performed with respect to time, so to calculate power, we need a way to record the time element. In the previous barbell press example, this is the duration of the concentric (upward) phase over which the work was performed—in that case, 0.815 seconds. The power achieved by pressing a 60-kilogram barbell 0.541 meters overhead in 0.815 seconds is 390.71 watts. And again, we can manipulate this by either performing more work in the same time or by performing the same work in less time (to increase power).

Over the years, developments in technology have enabled simultaneous recording of force and movement characteristics such as displacement and velocity. This has enabled researchers and practitioners to more easily calculate biomechanically derived performance metrics such as work and power, and while time-efficient performance tracking processes are critical, it can mean that important theoretical underpinnings are lost. Where it is possible to simultaneously record the force causing the velocity of a movement, power can be calculated by simply multiplying these two values. In doing this, the calculation of work and the time taken to perform it can be skipped entirely. This is a missed opportunity to gain the fullest understanding of the athlete's capacity at that point in time because this is information that could help inform what the athlete needs to do to improve their power capacity.

The ability to simultaneously record force and movement characteristics such as displacement and velocity has also led to the measurement, calculation, and use of **peak power**, which is frequently referred to as the *highest instantaneous power*. While this can result in some impressive numbers, the fact that it refers to the rate of work performed over one data sampling window is frequently

ignored, so its biomechanical relevance is misunderstood. For example, many systems record data at 1,000 hertz, which means data are recorded every 1 millisecond (ms). Is the rate that work is performed over 1 millisecond relevant? Caution is urged when considering this approach, and unless there is a specific need to know the rate of work over a single sample, it should be ignored. For example, the maximum instantaneous power in the previous barbell overhead press example was 642.63 watts, which describes the work performed in moving the barbell 1 millimeter. Nowhere is this method more popular than in vertical jump research. Going back to the example of the 74-kilogram, 40-centimeter jumper, the work the athlete performs during the propulsion phase of the jump (from the lowest squat to takeoff) is 295.57 joules, the athlete performs this over 0.315 seconds, and so achieves 938.31 watts of power. The peak power achieved during the same propulsion phase was 2040.68 watts, where the work performed involved 2 millimeters of displacement (of the jumper's center of mass).

We must mention an important limitation of using work and power to quantify movement performance. Both of these are **scalar** quantities, which means that they only provide a magnitude (how much) and cannot consider movement direction. Considering the importance of movement direction in sport and occupational performance, work and power should be used with this limitation in mind. Requiring nothing more than a tape measure and patience, practitioners can use work to quantify training load.

Negative Work and Power

When both force and movement are positive, both work and power will be positive too. In the earlier barbell pressing example, the lifter applies a positive force to push the barbell upward, so both force and movement are positive. However, while during the lowering phase of the exercise the force remains positive (to control the barbell's descent), its movement is negative (because it is traveling downward). We can use equation 2.3 to calculate the work

performed but this time replacing the positive displacement value (0.541 m) with a negative displacement value (-0.541 m). Because the only thing that has changed is movement direction, and because the range of motion should be the same, the lowering phase work is -318.43 J. This means that the power achieved during the lowering phase will be negative too. So far, we have considered how the application of force enables us to perform linear movement and work (movement over straight and curved paths). In the next section, we will explore how force applied to an object or body part results in rotational movement, such as when a segment of the body (e.g., the forearm) rotates around a joint like the elbow.

Relating Angular and Linear Motion

In biomechanics, we call body parts starting at a joint (e.g., the lower arm from the elbow) *segments*. If we flex the elbow to curl a dumbbell, the lower arm moves closer to the upper arm and the elbow angle decreases. The opposite is true when we lower the dumbbell. How far the dumbbell moves depends on the lower-arm length and change in elbow angle. If someone with a 0.32 meters-long (1.05 ft) lower arm flexes the elbow by 1.88 radians (about 108 degrees), the dumbbell will move 0.60 meters (1.97 ft). Increasing the elbow flexion to 2.27 radians (about 130 degrees) increases the dumbbell movement to 0.73 meters (2.40 ft). Try this with different segment lengths and joint angular displacements.

Rotation Force and Movement

Because the body is a system of levers (see Levers of the Musculoskeletal System), most movement is borne from the rotation of different parts of the body, all frequently contributing to more general movement. We use slightly different conventions to describe rotation (or angular movement), the forces that cause it, and its results. When the force of a contracting muscle (or muscles) causes the movement of a body segment, such as the forearm as it rotates around the elbow, we can quantify this by multiplying the force required to cause the movement by the length of the forearm and the sine of the joint angle (e.g., elbow, [figure 2.4, page 30](#)). The result is a torque (τ), although *moment* is frequently used

interchangeably. The subsequent movement is **angular displacement** (θ) which is measured in radians, and the rate of angular displacement is **angular velocity** (ω). While we should use the correct measurement (SI) units if we want to perform certain calculations with our data (e.g., convert linear motion to angular motion), measurement units like the radian can seem quite abstract, so they are frequently converted to degrees, which are often easier to conceptualize. To convert radians to degrees, we can use the calculation shown in equation 2.6, multiplying measurements in radians by 57.3. On the matter of measurement units, while we saw earlier that work can be reported in newton-meters ($\text{N} \cdot \text{m}$), it is also the measurement unit for torque and the one we will use throughout this section.

$$1 \text{ radian} = 180 \text{ degrees} \div \pi = 57.3 \quad (2.6)$$

Consider the dumbbell curl exercise where forearm length is 0.30 meters (0.98 ft), the dumbbell is 25 kilograms (55 lb), and the lifter is able to keep the upper arm tight against the body, preventing it from moving as he or she curls it through a vertical displacement of 0.60 meters (1.97 ft) in 1.283 seconds. The (linear) work performed by the lifter would be 140.64 joules at a linear power output of 109.59 watts. The angular displacement is 2.02 radians (or approximately 116 degrees), and the average angular velocity (Δ joint angle $\div$ time) is 1.57 radians per second (or 90 degrees per second). The average torque necessary to displace the dumbbell through that range of motion is 59.1 newton-meters, although this varies depending on the elbow joint angle where 27.7 newton-meters is required at the start (43 degrees), which gradually increases to 74.3 newton-meters when the elbow joint is at 90 degrees and then decreases to 50.2 newton-meters as the angle decreases to 43 degrees at the end of the concentric phase. We can calculate the required torque at a given joint angle by first multiplying the mass of the dumbbell by the acceleration of gravity and then multiplying the length of the forearm (from elbow center to dumbbell center) by the sine of the joint angle. At the start angle of

2.76 radians (158 degrees) and with the above information, the torque required to start the curl can be calculated using equation 2.7:

$$(25 \text{ kg} \cdot g) \times 0.30 \text{ m (elbow to dumbbell center)} \times \sin(2.76 \text{ radians}) = 27.4 \text{ N} \cdot \text{m} \quad (2.7)$$

This means that 124.3 joules of angular work is performed over 1.283 seconds at an angular power output of 96.9 watts. As earlier, we can manipulate the angular work that is performed by changing the mass of the object being moved, the range of motion, or both. With angular power, time provides another element that can be manipulated, because performing more work does not necessarily mean a higher power output unless any increase in performance duration is minimized.

Impulse also has a rotational equivalent (angular impulse), which is the product of torque and time ($\tau \times \Delta t$, where τ refers to torque, and Δt refers to the duration of the phase of interest). As with the linear impulse–momentum relationship, angular impulse is directly proportional to the change in angular momentum (ΔL), and because mass tends to remain constant, we can simplify this to angular impulse being proportional to a change in angular velocity ($\Delta \omega$). As with linear impulse, to move (rotate) faster, more (angular) impulse must be applied, and we know that this can be achieved by applying more torque, applying the same torque for longer, or a combination of both.

Strength Versus Power

The discrepancy between the common and scientific definitions of power has led to misunderstandings. For example, in the sport of powerlifting, which involves high forces but relatively low movement speeds, less mechanical power is produced than in several other sports, including weightlifting (8, 14). Despite the discrepancy, the sport of powerlifting is unlikely to be renamed. In all other contexts, the strength and conditioning professional should use the word *power* only in its scientific sense to avoid ambiguity and instead

consider *movement velocity*, where how quickly a movement is performed is of interest. Furthermore, although the word *strength* is often associated with slow velocities and the word *power* with high velocities of movement, both variables reflect the ability to exert force at a given velocity. Power is a direct mathematical function of force and velocity (which is why it becomes a scalar quantity). Earlier we saw that impulse is directly proportional to a change in velocity, providing a cause and effect. Therefore, it is not correct to associate strength with low speed and power with high speed. Strength is the capacity to exert force, and the amount of force we can apply over given time periods dictates how quickly the object this force is being applied to will move. If we need to move the object more quickly, we have to apply more impulse, and the constituent parts of the impulse equation that we change will be dictated by task demands and performance constraints (e.g., time). Where time is constrained, this means applying more force to balance this, and then if we need to move faster but cannot change the time constraint, this will mean developing the capacity to apply even more force during this time period. What is critical is the ability to exert force at velocities characteristic of a given sport to overcome gravity and accelerate the body or an implement. For a sport movement made relatively slow by high resistance, low-velocity strength is critical, whereas for a movement that is very fast due to low resistance, high-velocity strength is important. For example, when offensive and defensive American football linemen push against each other, their velocity of movement is slowed by the muscular force exerted by the opposing player as well as the inertia of the opposing player's body mass. Because the muscles are prevented from contracting at high velocity, the ability to exert force and power at low velocity is an important component of performance. In contrast, a badminton athlete's muscles quickly reach high velocity as a result of the minimal inertial resistance of the lightweight racket and the athlete's arm. Therefore, the ability to

exert force and power at high velocity is critical to making rapid adjustments in a stroke.

Key phases of the weightlifting movements are constrained by time because of the way the lifter interacts with the barbell. In powerlifting this is not the case, which is reflected in the impulse required to displace the barbell. The time constraints of weightlifting require the application of more force (relative to the barbell mass) in less time.

Biomechanical Factors in Human Strength

Several biomechanical factors are involved in the manifestation of human strength, including neural control, muscle cross-sectional area, muscle fiber arrangement, muscle length, joint angle, muscle contraction velocity, joint angular velocity, and body size. These factors are discussed next, as are the three-dimensional strength relationship and the strength-to-mass ratio.

Neural Control

Neural control affects the maximal force output of a muscle by determining which and how many motor units are involved in a muscle contraction (**recruitment**) and the rate at which the motor units are fired (**rate coding**) (6). Generally, muscle force is greater when more motor units are involved in a contraction, the motor units are greater in size, or the rate of firing is faster. Much of the improvement in strength evidenced in the first few weeks of resistance training is attributable to neural adaptations as the brain learns how to generate more force from a given amount of contractile tissue (44). It is not unusual for novice resistance trainees to become discouraged when they cannot maintain the rate of increase of the first few training weeks. It is important for them to realize that continued improvement will result if they adhere to the

training regimen, although via slower mechanisms such as muscle hypertrophy.

Muscle Cross-Sectional Area

All else being equal, the force a muscle can exert is related to its cross-sectional area rather than to its volume (15, 41). For example, if two athletes of similar percent body fat but different height have the same biceps circumference, their upper arm muscle cross-sectional areas are about the same. Although the taller (and therefore heavier) athlete's longer muscle makes for greater muscle volume, the strength of the two athletes' biceps should be about the same. With the same strength but greater body weight, the taller athlete has less ability to lift and accelerate his or her own body—for example, when performing calisthenics or gymnastics. This is why most elite gymnasts are not very tall. As described in [chapter 1](#), resistance training increases both the strength and cross-sectional area of muscle.

Arrangement of Muscle Fibers

Maximally contracting muscles have been found capable of generating forces of 23 to 145 pounds per square inch (psi) (16-100 N/cm²) of muscle cross-sectional area (28). This wide range can be partially accounted for by the variation in the arrangement and alignment of sarcomeres in relation to the long axis of the muscle ([figure 2.11](#)) (28). A **pennate muscle** has fibers that align obliquely with the tendon, creating a featherlike arrangement. The **angle of pennation** is defined as the angle between the muscle fibers and an imaginary line between the muscle's origin and insertion; 0 degrees corresponds to no pennation.

Many human muscles are pennated (27, 51, 53), but few have angles of pennation in excess of 15 degrees. Actually, the angle of pennation does not remain constant for a given muscle but increases as the muscle shortens. Any factor that affects angle of pennation would thus affect strength and velocity of shortening as long as the

cross-sectional area remains the same. Muscles with greater pennation have more sarcomeres in parallel and fewer sarcomeres in series; they are therefore better able to generate force but have a lower maximal shortening velocity than non-pennate muscles. In comparison, lesser amounts of pennation can be advantageous for producing high velocities due to the greater number of sarcomeres in a row, at the expense of number of sarcomeres in parallel. The amount of pennation, however, has an effect on the muscles' ability to generate eccentric, isometric, or low-speed concentric force (60). Most importantly, although angle of pennation may vary depending on hereditary factors, it is modifiable through training, which could help account for some of the differences in strength and speed seen in individuals who seem to have muscles of the same size.

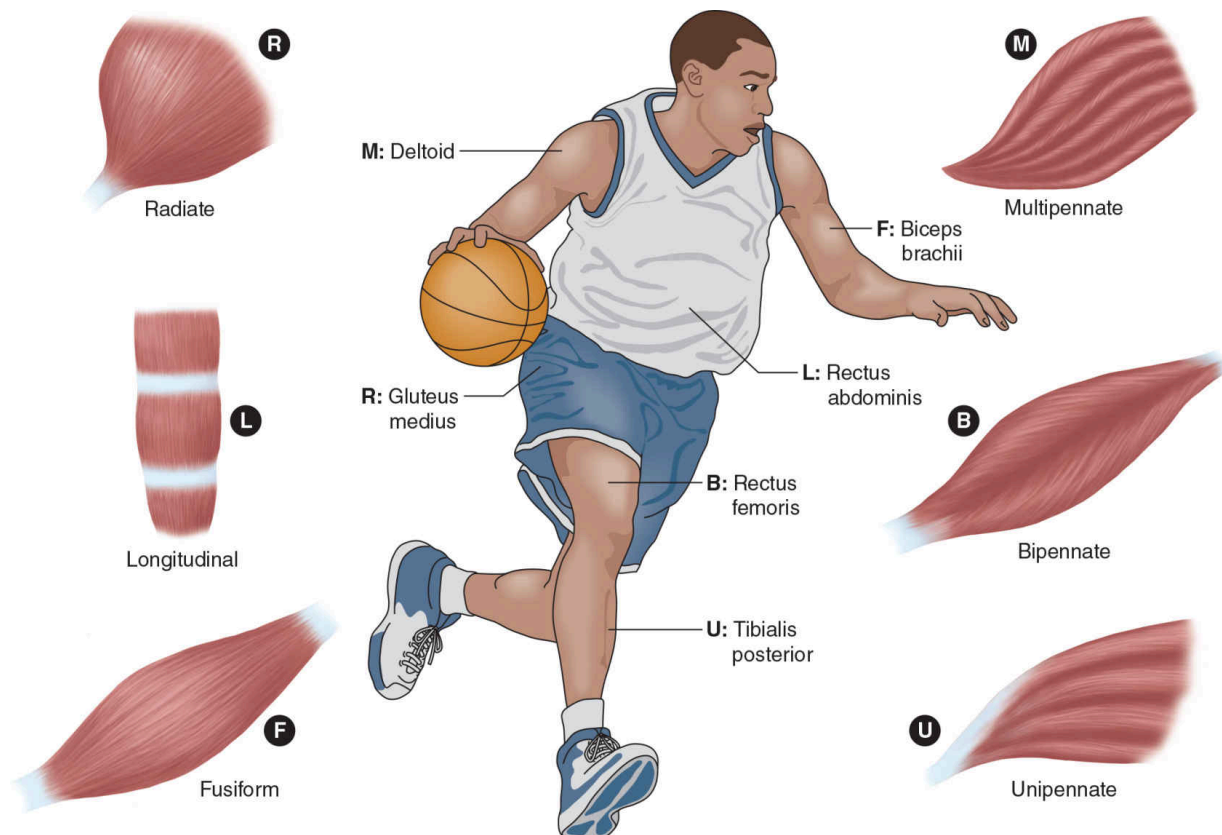


FIGURE 2.11 Muscle fiber arrangements and an example of each.

Muscle Length

At rest, actin and myosin filaments in muscle fibers lie alongside each other, allowing for a maximal number of potential crossbridge sites ([figure 2.12](#); see [chapter 1](#) for more details). This alignment enables the muscle to generate the greatest force at its resting length. When the muscle is stretched beyond its resting length, fewer actin and myosin filaments align next to each other, reducing the number of potential crossbridge sites and, consequently, the force the muscle can generate. In this stretched state, a third filament, **titin**, is recognized to play a critical role in force development. Titin not only contributes to structural alignment but also possesses mechanosensing properties. During eccentric muscle actions beyond resting length, the increase in calcium concentration in the sarcoplasm causes the spring-like titin to tighten. This tightening helps in force production and aligns myosin filaments to increase the potential crossbridge sites (23, 24, 61). Under load, titin is considered to be a key influencer in trophic signaling, which can be a contributor to skeletal muscle hypertrophy (26). Conversely, when the muscle contracts too much below its resting length, the actin filaments overlap excessively, reducing the number of crossbridge sites and thus decreasing force generation capability.

Joint Angle

Because all body movements, even those occurring in a straight line, take place by means of rotation about a joint or joints, the forces that muscles produce must be manifested as torques; recall that a higher torque value indicates a greater tendency for the applied force to rotate the limb or body part about a joint. Consequently, we speak of torque versus joint angle rather than force versus joint angle. The amount of torque that can be exerted about a given body joint varies throughout the joint's range of motion, largely because of the relationship of force versus muscle length, as well as the ever-changing leverage brought about by the dynamic geometry of the muscles, tendons, and internal joint structures. This is shown in

[figures 2.2, 2.3, and 2.4](#) ([pages 29-30](#)). Additional factors include the type of exercise (isotonic, isometric, and so on), the body joint in question, the muscles used at that joint, and the speed of contraction (13).

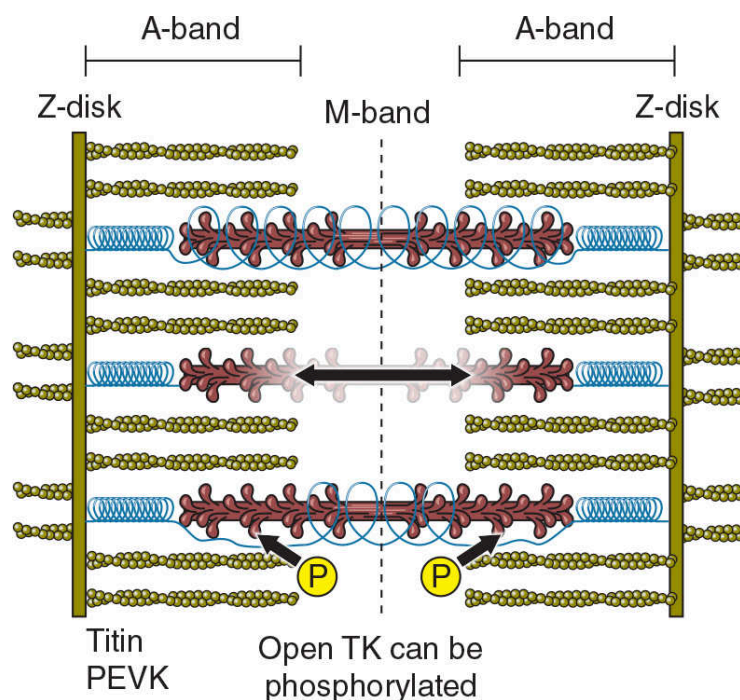


FIGURE 2.12 A schematic of the interaction between actin, myosin, and titin filaments when the muscle is at its resting length and when it is contracted or stretched. Muscle force capability is greatest when the muscle is at its resting length because of increased opportunity for actin–myosin crossbridges, while titin contributes to force production and alignment during eccentric muscle actions beyond resting length.

Muscle Contraction Velocity

Classic experiments by A.V. Hill (25, 35) on isolated animal muscle showed that the force capability of muscle declines as the velocity of contraction increases. The relationship is not linear; the decline in force capability is steepest over the lower range of movement speeds ([figure 2.13](#); 30). Human movement technique can make the best of this relationship. For example, as a vertical jump begins, the arms swing upward, thereby exerting downward force on the body at the shoulders, slowing the upward movement of the body, and forcing the hip and knee extensor muscles to contract more slowly

than they otherwise would, enabling them to generate higher forces for longer times.

Joint Angular Velocity

There are three basic types of muscle action, during which forces are generated within the muscle that pull the muscle's ends toward each other if not prevented from doing so by external forces. The term *muscle action* is preferable to *contraction*, because the latter means "shortening," which does not accurately describe two of the three muscle actions.

- In a **concentric muscle action**, the muscle shortens because the contractile force is greater than the resistive force. The forces generated within the muscle and acting to shorten it are greater than the external forces acting at its tendons to stretch it. Swimming and cycling involve concentric muscle action almost exclusively.
- In an **eccentric muscle action**, the muscle lengthens because the contractile force is less than the resistive force. The forces generated within the muscle and acting to shorten it are less than the external forces acting at its tendons to stretch it (which increases the risk of soreness and injury). This occurs during the lowering phase of any resistance exercise. During standard resistance training, the eccentric force exerted by the muscle keeps the weight from being accelerated downward by gravitational force. Thus, the weight moves steadily downward rather than picking up speed and contacting the floor or the athlete's body.
- In an **isometric muscle action**, the muscle length does not change, because the contractile force is equal to the resistive force. The forces generated within the muscle and acting to shorten it are equal to the external forces acting at its tendons to stretch it. During a sit-up with the trunk held straight, the abdominal muscles act isometrically to maintain the rigidity of the trunk, while the hip flexors carry out the sit-up movement.

In contrast, the abdominal muscles act concentrically and eccentrically during the raising and lowering phases of the curl-up exercise, respectively.

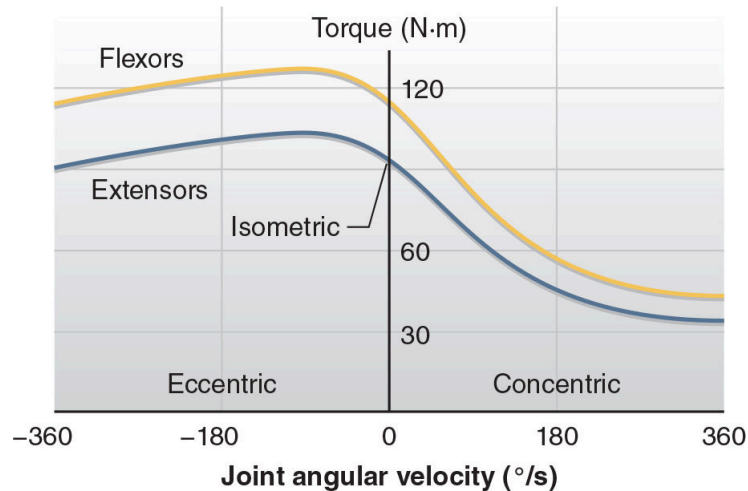


FIGURE 2.13 Force–velocity curve for eccentric and concentric actions.

Reprinted from K. Jorgensen, "Force-Velocity Relationship in Human Elbow Flexors and Extensors," in *Biomechanics A-V*, edited by P.V. Komi, (University Park Press, 1976), 147. By permission of P.V. Komi.

Muscle torque varies with joint angular velocity according to the type of muscular action ([figure 2.13](#)). Tests have shown that during isokinetic (constant-speed) concentric exercise by human subjects, torque capability declines as angular velocity increases. In contrast, during eccentric exercise, as joint angular velocity increases, maximal torque capability increases until about 90 degrees per second (1.57 rad/s), after which it declines gradually (6). That means that the greatest muscle force can be obtained during eccentric muscle action. This is demonstrated by athletes who employ "cheating" movements when a weight cannot be lifted using strict form. For example, an individual who reaches a **sticking point** in the biceps curl exercise due to the limit of concentric elbow flexor strength usually leans the torso back, allowing the elbow flexors to exert increased force by operating isometrically or eccentrically and thereby enabling continued movement of the bar. This sticking point is represented by a decrease in acceleration.

Strength-to-Mass Ratio

In sport activities such as sprinting and jumping, the ratio of the strength of the muscles involved in the movement to the mass of the body parts being accelerated is critical. Thus, the strength-to-mass ratio directly reflects an athlete's ability to accelerate his or her body. If, after training, an athlete increases body mass by 15% but increases force capability by only 10%, the strength-to-mass ratio, and thus the athlete's ability to accelerate, is reduced. A sprinter or jumper may benefit by experimenting with muscle mass to determine the highest strength-to-mass ratio, which would result in the best possible performance.

In sports involving weight classification, the strength-to-mass ratio is extremely important. If all competitors have close to the same body mass, the strongest one has a decided advantage. It is normal for the strength-to-mass ratio of larger athletes to be lower than that of smaller athletes because when body size increases, muscle volume (and concomitantly body weight) increases proportionately more than does muscle cross-sectional area (and concomitantly strength) (12). Trial and error can help athletes determine the weight category in which their strength is highest relative to that of other athletes in the weight class. Once an athlete finds his or her most competitive weight class, the object is to become as strong as possible without exceeding the class weight limit.

Body Size

It has long been observed that, all else being equal, smaller athletes are stronger pound for pound than larger athletes (12). The reason is that a muscle's maximal contractile force is fairly proportional to its cross-sectional area, which is related to the square (second power) of linear body dimensions, whereas a muscle's mass is proportional to its volume, which is related to the cube (third power) of linear body dimensions. Therefore, as body size increases, body mass increases more rapidly than does muscular strength. Given constant

body proportions, the smaller athlete has a higher strength-to-mass ratio than does the larger athlete (12).

There has always been interest in comparing the performances of athletes in different weight categories. The most obvious method for doing so is to divide the weight lifted by the athlete's body weight. However, such an adjustment is biased against larger athletes because it does not consider the expected drop in the strength-to-mass ratio with increasing body size. Various formulas have been derived to more equitably compare loads lifted. In the **classic formula**, the load lifted is divided by body weight to the two-thirds power, thus accounting for the relationship of cross-sectional area versus volume. Other formulas have since been developed because the classic formula seemed to favor athletes of middle body weight over lighter and heavier athletes (7). However, the determination by the classic formula that the performances of medium-weight athletes are usually the best may indeed be unbiased. Because of the bell-shaped curve describing the normal distribution of anthropometric characteristics among the population, the body weights of a majority of people are clustered close to the mean.

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